

STUDENT PERFORMANCE ANALYSIS

Executive Report | Education Domain

STUDENTS	1,200
AVERAGE SCORE	66.5
PASS RATE	93.8%
AT-RISK STUDENTS	321 (26.8%)

Executive Summary

The analysis covers 1,200 cleaned student records. Overall average performance is 66.5, with an all-subject pass rate of 93.8%. Computer is the strongest subject (69.8), while Mathematics requires the greatest academic support (62.4).

Attendance has a correlation of 0.454 with average score, while daily study hours correlate at 0.677. These associations support early outreach and targeted tutoring, but do not establish causation.

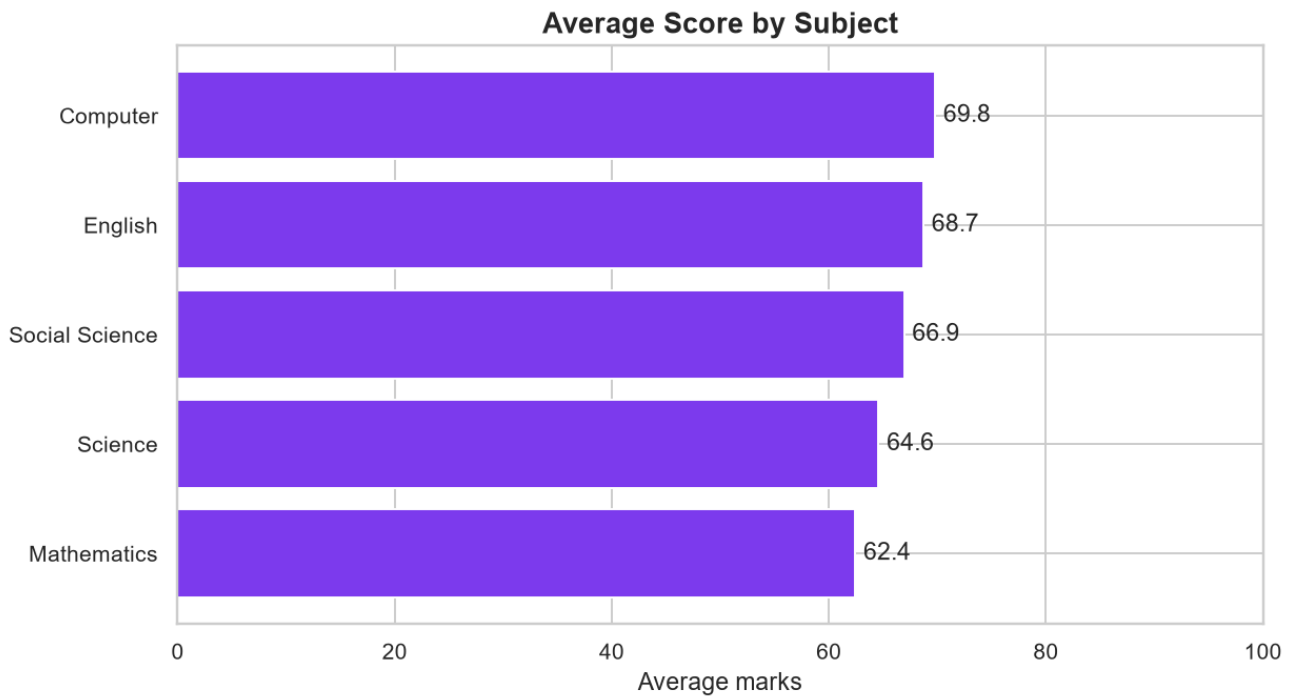
1. Methodology and Definitions

Records were checked for schema completeness, duplicate IDs, missing data, and valid ranges. Missing scores were imputed using grade-level medians. Student averages, medians, subject passes, overall result, performance band, and risk status were engineered.

A subject is passed at 40 marks. A student passes overall only when all five subjects are passed. At-risk status is assigned when attendance is below 75%, any subject is failed, or the overall average is below 50.

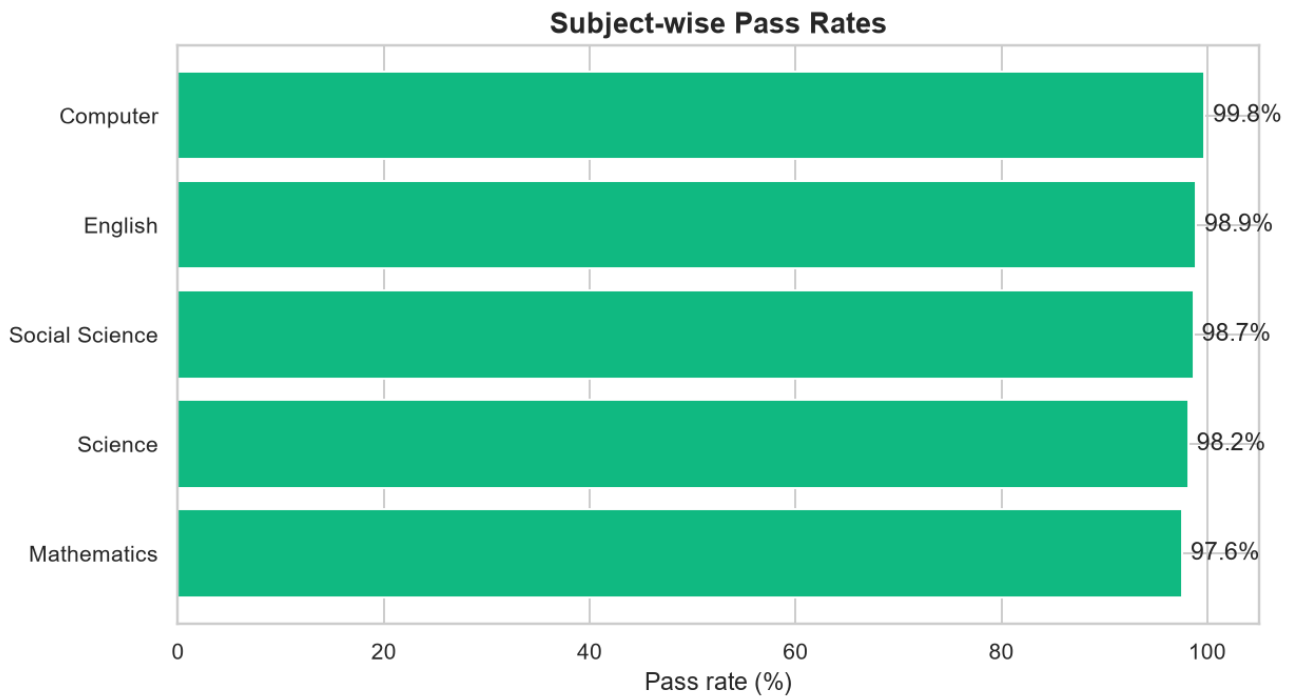
Metric	Value
Overall median	66.18
Average attendance	81.72%
Attendance-score correlation	0.454
Study-hours correlation	0.677

2. Subject Performance



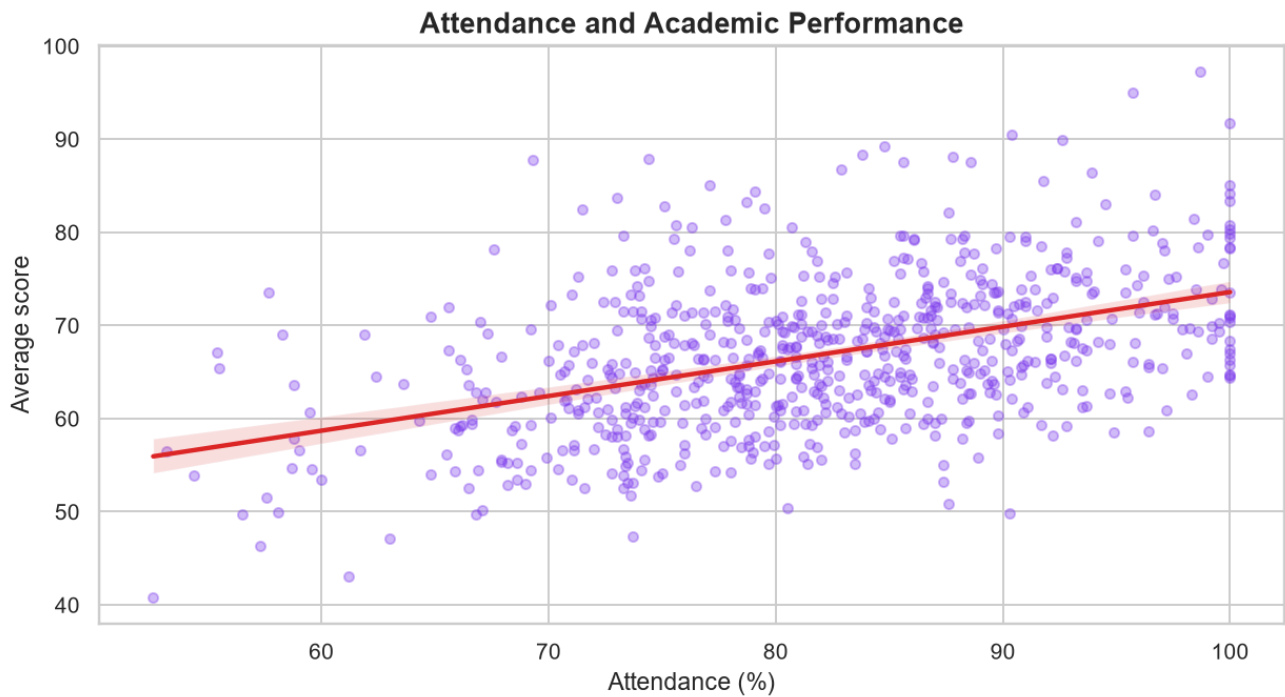
Computer leads average performance; Mathematics should receive targeted curriculum and tutoring review.

3. Subject Pass Rates



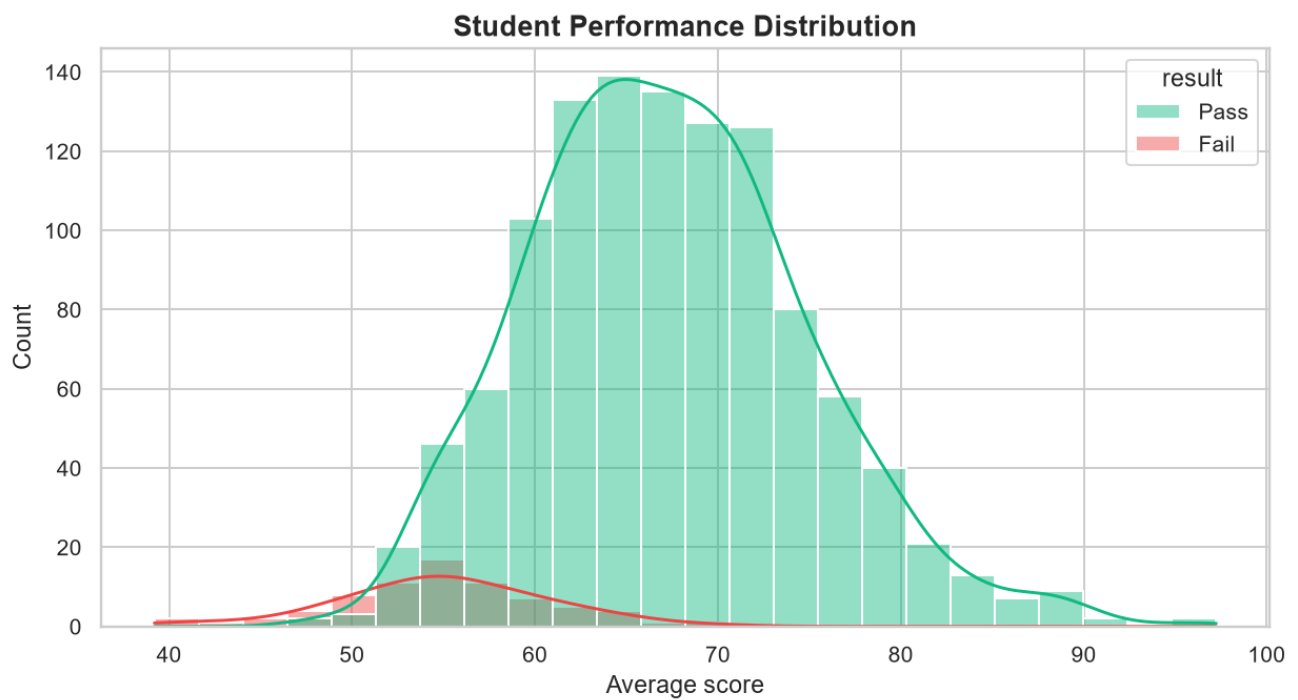
Subject pass rates reveal where average marks may hide a meaningful lower-performing group.

4. Attendance Impact



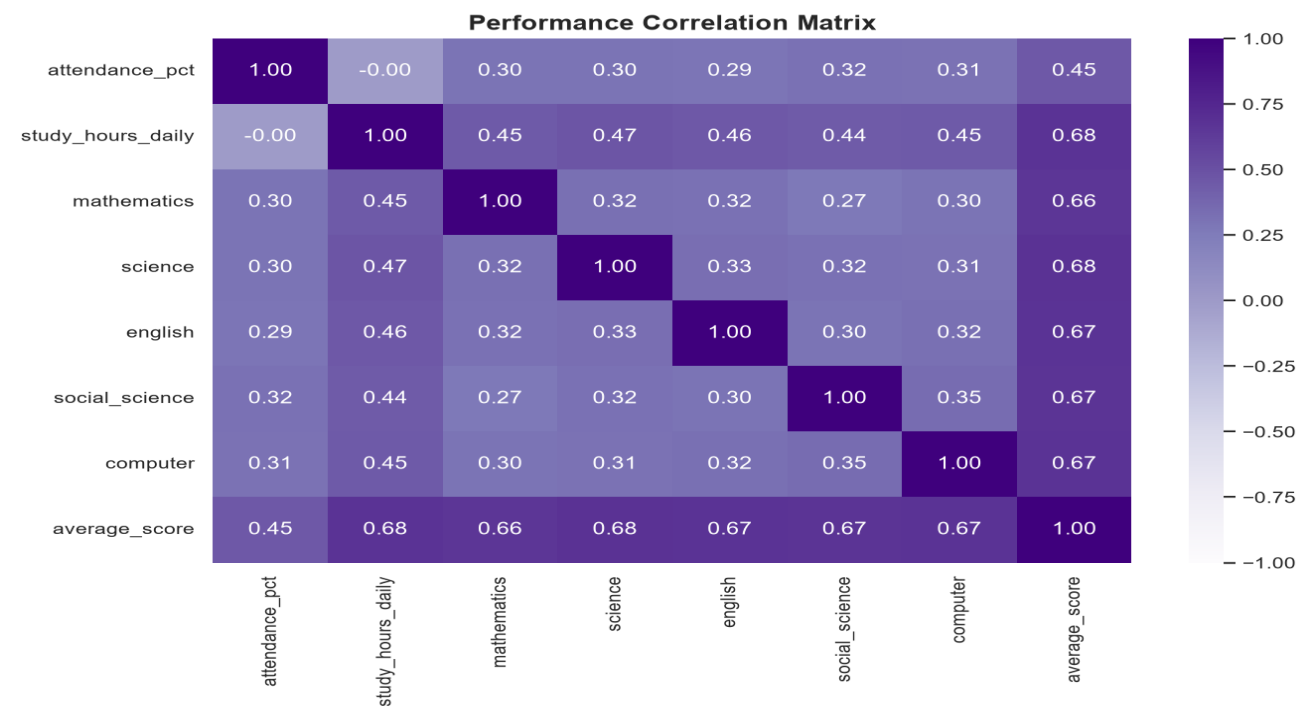
The fitted line shows the direction of the attendance-performance relationship. It is an association, not proof that attendance alone causes higher scores.

5. Score Distribution



The distribution compares passing and failing students and highlights the score range where intervention may have the greatest impact.

6. Correlation Analysis



The matrix compares academic and behavioural variables, helping prioritise factors for further investigation.

7. Recommendations and Safeguards

Attendance alerts: Contact students before attendance falls below 75%, focusing on barriers and support.

Targeted tutoring: Prioritise Mathematics and students failing individual subjects.

Progress reviews: Review each at-risk student's attendance, subject scores, and study plan every two weeks.

Access support: Offer supervised computer and library access where home internet is limited.

Evaluation: Measure attendance improvement, subject gain, pass conversion, and risk-exit rate.

Ethical note

Risk flags are screening aids, not final judgements. Teachers should review context, avoid punitive use, protect privacy, and monitor whether the rule affects groups unfairly. The portfolio dataset is simulated and contains no real personal information.